

University of Dar es Salaam
Department of Mathematics
MT233: Mathematical Statistics, 2024/2025

Tutorial 1

1. If $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$, then prove that $P(A \cap B) \geq \frac{3}{8}$
2. Suppose $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.15$. Find the $P(A/\bar{B})$
3. Three students A , B and C are solving the same problem in a given test. The probabilities that A , B and C will solve the problem are $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ respectively. What is the probability that the problem is solved?
4. In a shooting test, the probability of hitting the target is $\frac{1}{2}$ for A , $\frac{2}{3}$ for B and $\frac{3}{4}$ for C . If all of them fire at the target, find the probability that
 - (a) none of them hits the target
 - (b) at-least one of them hits the target
5. Three candidates A , B and C are selected for the position of a general manager in a company whose chances of getting the appointment are in the proportion $4 : 2 : 3$ respectively. The probability that if A is selected will improve the company performance is 0.3. The probability of B and C doing the same are 0.5 and 0.8 respectively. What is the probability that the company performance will be improved?
6. Suppose that, in a particular city, airport A handles 50% of all airline traffic, and airports B and C handle 30% and 20%, respectively. The detection rates for weapons at the three airports are 0.9, 0.8, and 0.85, respectively. If a passenger at one of the airports is found to be carrying a weapon through the boarding gate, what is the probability that the passenger is using (a) airport A ? (b) Airport C ?
7. There 4 men and 5 women in a surveying office. The customer wants a site visit. What is the probability of a customer getting site visit from a group of 2 men and 2 women?